

Manual

HYDRA Interfacing Module to SAP PP-PI SAP-PPPI 8.2

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1 HYDRA Interface to SAP PP-PI via PP-PI-PCS

Fields of Application

The interface HKMPP-PI has been designed to connect the MES system HYDRA to superior SAP systems when SAP PP-PI is used on the ERP level.

Implementation Notes

The SAP-PPPI interface is used, when

- SAP PP-PI is used to manage production using SAP process orders
- Phase confirmation is used

Integration

If you use the SAP-PPPI component, the added orders/ operations form the reporting structure for numerous additional components in HYDRA.

Features

- Reception of process order data using the mechanisms that SAP provides related to control recipes
- Transfer of phase confirmation using the PP-PI-PCS standard structure PI_PHCON

2 Data type definitions

Type	Description
CHAR x	Information is left-aligned for the data type CHAR. Places that are not required are filled with blanks (U+0020). If the field is not used, it must be completely prepopulated with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. The data type NUMC only supports digits (ASCII characters 30 Hex to 39 Hex). These digits are right-aligned and unnecessary places are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030). Example: "00000002"
DEC x.y QUAN x.y	Numeric field of the length x and y decimal places. A data field in HYDRA format is preceded by a sign ("+" or "-") and includes a decimal point. Places that are not required are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030) including algebraic sign and decimal separator. e.g. DEC 13,3: • -1234567890,123 → -1234567890.123 • 234567890,3 → +0234567890.300 Note: The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. This means, for example, that a field QUAN 13.3 is converted to an external length of CHAR15.
DATE	Format YYYYMMDD. If the field is not used, it must remain empty (filled with blanks (U+0020)).
TIME	Format HHMMSS. If the field is not used, it must be set to "000000" (zeros with (U+0030)).



HYDRA does not support special characters for all alphanumeric fields. This includes, among others: "\" (backslash), "|" (pipe), „“““ (double quotes), and " ’ “ (single quotes). You cannot enter these characters using shop floor terminals and the MOC does not support them.



The character " ; " (semicolon) is used as a separator in the system. You must not use this character in key fields (e.g. order/operation number, MES batch number, personnel number).



The character " % " (percent) is used as a placeholder for database communication. You should not use this character to prevent the result from being falsified.

3 Process Order data SAP --> HYDRA

General remarks

To download the process order related data from SAP to HYDRA the SAP standard interface PP-PI-PCS is used. That interface can be prosecuted in different modes. In the implementation of HYDRA in the scope of the HKMPP-PI interface the mode used is “Download to ext. system, initiated by SAP process management“. That means that a control recipe destination of type “2” has to be created in SAP customizing. By that the download of process order data will be initiated from SAP only.

In recepies in SAP operations and phases can be created. For the download to HYDRA and the recording in HYDRA only phases are of interest. That means that only the phases have to be linked to a control recipe receiver in the master recipe. The phases in SAP are treated in HYDRA as operation. The length of the key for the order number and the operation number can be configured in HYDRA. For details on application relevant settings in SAP please see chapter [Application-relevant customizing in SAP](#).

In the following chapters the structure of the process order download will be described including the used categories and its characteristics. It is essential to understand, that the order of the characteristics is essential for HYDRA. All characteristics described here have to be created and customized in SAP in the exact order (while the name used in SAP to identify it, does not matter to HYDRA).

If in later steps the categories have to be extended, that only is allowed at the end of the category.

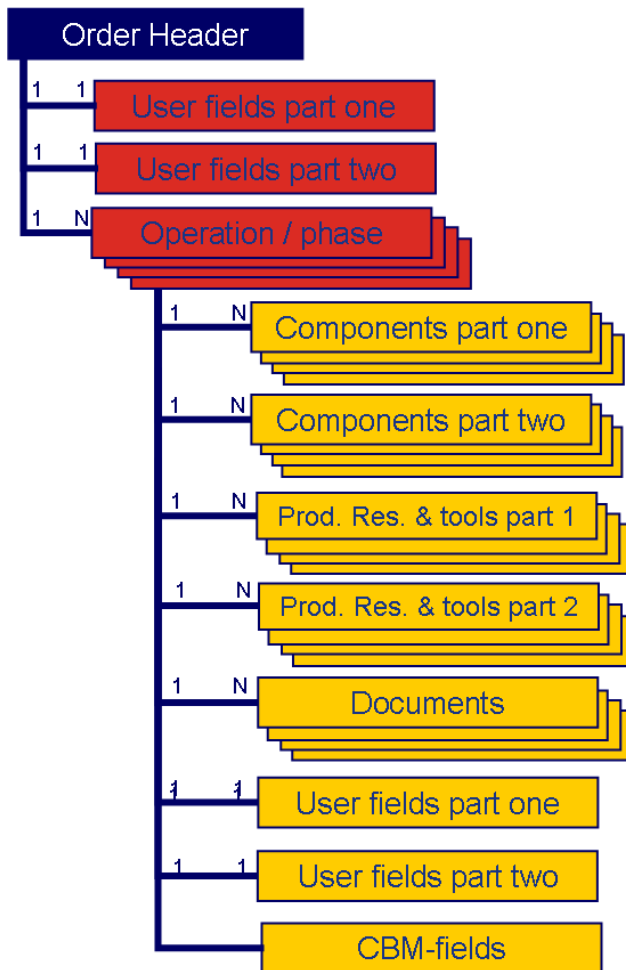
Download of process orders

The process orders are created in SAP in advance. In SAP they then have the status “Created“. The PP-PI-PCS interface does not support the download of process orders that in SAP have system status “CREATED“. By releasing the process order in SAP (automatically / manually), depending on the setting in the production scheduling profile the control recipe is created as well or needs to be created manually. The download of the control recipe with the data of the process order from SAP to HYDRA is initiated from SAP either manually or by a scheduled job. For details of application relevant settings in SAP please see chapter [Application-relevant customizing in SAP](#).

Deletion of process orders

It might happen that process orders are canceled in SAP after the transfer to HYDRA. In that case the control recipe is discarded in SAP and the process order is canceled. As SAP PP-PI is not able to deliver this information, the deletion / closing of the process order has to be done manually in HYDRA.

Interface structure



Message type)*: PP_PI_PCS_HYDRA_INBOUND

IDoc type)*: PP_PI_PCS_HYDRA_INBOUND

Message functions: -

Segments	ZHEAD_01 (order header)	1-time
/ categories:	└ ZUFIELD1 Userfield part one	1-time
	└ ZUFIELD2 User fields part two	1-time
	└ ZOPER_01 Operation data part one	n-times
	└ ZOPER_02 Operation data part two	n-times
	└└ ZCOMP_01 Components list part one	n-times
	└└ ZCOMP_02 Components list part two	n-times
	└└ ZAGFH_01 Production resources and tools/resources	n-times
	└└ ZAGFH_02 Production resources and tools/resources	n-times
	└└ ZAGDC_01 Documents	n-times
	└└ ZUFIELD3 User fields part one	1-time
	└└ ZUFIELD4 User fields part two	1-time
	└└ ZAGRF_02 coil based manufacturing	1-time

)* The information is used in HYDRA only for HYDRA-internal reasons

The structures belonging to the single segments are described as follows. The single columns have the following meaning:

Column	Meaning
Field	Designation of the field
V(use)	S It is a key field which identifies the data record clearly if needed together with other fields characterized as key fields. The field must be filled. M It is a mandatory field which has to be filled. ML Mandatory field if the control center of HYDRA is used (HLS). MM Mandatory field if the material and production logistics of HYDRA is used (MPL or MPL/RF). K Field is allowed to stay empty.
T(type)	Data type according to the description in the chapter above.
L(length)	Field length for fields of the data type DEC: total number of places, without decimal point and algebraic sign.
D(decimal places)	For fields of the data type DEC: number of places after decimal point, otherwise: not relevant
Description	Description of the field or comment to the field

Order Header (ZHEAD_01)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAUART	01-AUART	AUART	M	CHAR	5		Order type NOTE: The order types have to be customized in HYDRA. For that only specified order types are allowed to be transferred. If no project-sepcific definition is done, fixed "0"
0030	PPPI_MATERIAL_NUMBER	MARA-BISMT	ATK	M	CHAR	18		Producing material (Article); alphabetical characters in the UPPER CASE
0040	PPPI_MATERIAL_SHORT_TEXT	01-MATXT	ATKBEZ_1	K	CHAR	30		Designation of the article
0050	ZPPPI_MATERIAL_SHORT_TEXT		ATKBEZ_2	K	CHAR	10		Designation of the article
0060	ZPPPI_CUSTOMER_DESCRIPTION_01	01-KUNUM	KDBEZ_1	K	CHAR	30		Designation of the customer
0070	ZPPPI_CUSTOMER_DESCRIPTION_02		KDBEZ_2	K	CHAR	10		Designation of the customer
0080	ZPPPI_SALES_ORDER_NUMBER	01-KDAUF_AUFG	KDAUNR	K	CHAR	25		Customer order
0090	ZPPPI_SALES_ORDER_ITEM	01-KDPOS_AUFG	KDAUPOS	K	CHAR	15		Customer order position
0100	ZPPPI_PRIORITY	01-APRIO	EXTPRIO	M	CHAR	1		Priority
0110	ZPPPI_INDEX		AUIDX	K	DEC	5	2	Order index; should be 0.00 Information: According to the description of the data type it should be considered that an algebraic sign and a decimal point have to be supplemented to the length that is indicated here. This is valid for all following fields of the data type DEC.
0120	PPPI_UNIT_OF_MEASURE	01-GMEIN	SGE:B	M	CHAR	3		Base quantity unit
0130	PPPI_ORDER_QUANTITY	01-GAMNG	SGR:GUTB	M	DEC	13	3	Target quantity (base quantity unit)
0140	ZPPPI_TARGET_SCRAP		SGR:AUSB	K	DEC	13	3	Target scrap (base quantity unit)
0150	Z_PPPI_MATERIAL_TYPE_HYDRA		MATTYP	M	CHAR	20		Material type (configured in HYDRA) of the producing material (article)
0160	PPPI_BATCH		CNR	K	CHAR	20		Batch number, no processing in HYDRA
0170	PPPI_INSPECTION_LOT	01-PRUEFLOS	PCNR	K	CHAR	20		Inspection order/inspection lot number
0180	ZPPPI_SAMPLE_CATEGORY		PPKTYP	K	CHAR	1		Physical-sample category
0190	ZPPPI_EARLIEST_START_DATE	01-GSTRP	DATFB	ML	DATE	8		Earliest start (date) SAP: Start BasicDates (AFKO.GSTRP)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0200	Z_PPI_EARLIEST_START_TIME	01-GSUZP	ZEIFB	ML	TIME	6		Earliest start (time) SAP: Start BasicDates (AFKO.GSUZP) Please note: 240000 is not allowed; use instead 235959
0210	ZPPPI_LATEST_END_DATE	01-GLTRP	DATSE	ML	DATE	8		Latest end (date) SAP: Finish BasicDates (AFKO.GLTRP)
0220	Z_PPI_LATEST_END_TIME	01-GLUZP	ZEISE	ML	TIME	6		Latest end (time) SAP: Finish BasicDates (AFKO.GLUZP) Please note: 240000 is not allowed; use instead 235959
0230	ZPPPI_SCHEDULED_START	01-GSTRS	DATTERM	K	DATE	8		Scheduled start (date) (AFKO.GSTRS)
0240	Z_PPI_SCHEDULED_START_TIME	01-GSUZS	ZEITERM	K	TIME	6		Scheduled start (time) (AFKO.GSVZS) Please note: 240000 is not allowed; use instead 235959
0250	ZPPPI_SCHEDULED_FINISH	01-GLTRS	DATTERME	K	DATE	8		Scheduled end (date) (AFKO.GLTRS)
0260	Z_PPI_SCHEDULED_FINISH_TIME	01-GLUZS	ZEITERME	K	TIME	6		Scheduled (time) (AFKO.GLUZS) Please note: 240000 is not allowed; use instead 235959
0270	ZPPPI_SCHEDULED_TYPE	01-TERKZ	TERMART	K	CHAR	1		Scheduling type. Mandatory field if the scheduling is supposed to take place in HYDRA. V = Forward R = Backward
0280	ZPPPI_REDUCTION_STRATEGY	01-TERKZ	REDSTRAT	K	CHAR	2		à Reduction strategy. According to HYDRA Customizing
0290	ZPPPI_PRODUCTION_SCHEDULER	01-FEVR	AUGRP	K	CHAR	4		Order group belongs to "production scheduler" in SAP
0300	ZPPPI_MRP_CONTROLLER	01-DISPO	DISP	K	CHAR	10		MRP controller
0310	ZPPPI_LEADING_ORDER	01-LEAD_AUFNR	PRJNR	K	CHAR	25		Project number; belongs to the WBS number in SAP
0320	ZPPPI_SUPERIOR_ORDER	AFPO-PLNUM	PLANAUNR	K	CHAR	25		Planned order
0330	ZPPPI_COST_OBJECT		KTR	K	CHAR	25		Cost object
0340	ZAPNR_1	01-PLNNR	APNR_1	K	CHAR	30		Working plan
0350	ZAPNR_2		APNR_2	K	CHAR	10		Working plan

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0360	ZAPVER	01-PLNAL	APVER	K	CHAR	12		Working plan planned order version
0370	ZSLVER		SLVER	K	CHAR	12		Bill of material version
0380	ZKLKK:MNR		KLKK:MNR	K	DEC	13	3	Calculated costs machine
0390	ZKLKK:L		KLKK:L	K	DEC	13	3	Calculated costs wage
0400	ZKLKK:MAT		KLKK:MAT	K	DEC	13	3	Calculated costs material
0410	ZKLKK:SONST		KLKK:SONST	K	DEC	13	3	Calculated costs other
0420	ZMATWERT:GUT		MATWERT:GUT	K	DEC	13	3	Material value
0430	ZMATWERT:AUS		MATWERT:AUS	K	DEC	13	3	Scrap value
0440	PPPI_CONTROL_RECIPE		STRGREZ	K	CHAR	18		Control Recipe

Userfields of the order header

User fields part one (ZUFIELD1)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAAUUF_USRFLD		USRFLD	S	CHAR	8		User field key
0030	ZAAUUF_FU:1		FU:1	K	DATE	8		User field 1
0040	ZAAUUF_FU:2		FU:2	K	DATE	8		User field 2
0050	ZAAUUF_FU:3		FU:3	K	DATE	8		User field 3
0060	ZAAUUF_FU:4		FU:4	K	DATE	8		User field 4
0070	ZAAUUF_FU:5		FU:5	K	DATE	8		User field 5
0080	ZAAUUF_FU:6		FU:6	K	DATE	8		User field 6
0090	ZAAUUF_FU:7		FU:7	K	NUM	8		User field 7
0100	ZAAUUF_FU:8		FU:8	K	NUM	8		User field 8
0110	ZAAUUF_FU:9		FU:9	K	NUM	8		User field 9
0120	ZAAUUF_FU:10		FU:10	K	NUM	8		User field 10
0130	ZAAUUF_FU:11		FU:11	K	NUM	8		User field 11

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0140	ZAAUUF_FU:12		FU:12	K	NUM	8		User field 12
0150	ZAAUUF_FU:13		FU:13	K	NUM	8		User field 13
0160	ZAAUUF_FU:14		FU:14	K	NUM	8		User field 14
0170	ZAAUUF_FU:15		FU:15	K	NUM	8		User field 15
0180	ZAAUUF_FU:16		FU:16	K	NUM	8		User field 16
0190	ZAAUUF_FU:17		FU:17	K	NUM	8		User field 17
0200	ZAAUUF_FU:18		FU:18	K	NUM	8		User field 18
0210	ZAAUUF_FU:19		FU:19	K	NUM	8		User field 19
0220	ZAAUUF_FU:20		FU:20	K	NUM	8		User field 20
0230	ZAAUUF_FU:21		FU:21	K	NUM	8		User field 21
0240	ZAAUUF_FU:22		FU:22	K	NUM	8		User field 22
0250	ZAAUUF_FU:23		FU:23	K	DEC	13	3	User field 23
0260	ZAAUUF_FU:24		FU:24	K	DEC	13	3	User field 24
0270	ZAAUUF_FU:25		FU:25	K	DEC	13	3	User field 25
0280	ZAAUUF_FU:26		FU:26	K	DEC	13	3	User field 26
0290	ZAAUUF_FU:27		FU:27	K	DEC	13	3	User field 27
0300	ZAAUUF_FU:28		FU:28	K	DEC	13	3	User field 28
0310	ZAAUUF_FU:29		FU:29	K	CHAR	1		User field 29
0320	ZAAUUF_FU:30		FU:30	K	CHAR	1		User field 30
0330	ZAAUUF_FU:31		FU:31	K	CHAR	1		User field 31
0340	ZAAUUF_FU:32		FU:32	K	CHAR	1		User field 32
0350	ZAAUUF_FU:33		FU:33	K	CHAR	1		User field 33
0360	ZAAUUF_FU:34		FU:34	K	CHAR	1		User field 34
0370	ZAAUUF_FU:35		FU:35	K	CHAR	1		User field 35
0380	ZAAUUF_FU:36		FU:36	K	CHAR	1		User field 36
0390	ZAAUUF_FU:37		FU:37	K	CHAR	1		User field 37
0400	ZAAUUF_FU:38		FU:38	K	CHAR	1		User field 38
0410	ZAAUUF_FU:39		FU:39	K	CHAR	1		User field 39

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0420	ZAAUUF_FU:40		FU:40	K	CHAR	1		User field 40
0430	ZAAUUF_FU:41		FU:41	K	CHAR	1		User field 41
0440	ZAAUUF_FU:42		FU:42	K	CHAR	1		User field 42
0450	ZAAUUF_FU:43		FU:43	K	CHAR	1		User field 43
0460	ZAAUUF_FU:44		FU:44	K	CHAR	1		User field 44

User fields part two (ZUFIELD2)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAAUUF_USRFLD		USRFLD	S	CHAR	8		User field key
0030	ZAAUUF_FU:45		FU:45	K	CHAR	10		User field 45
0040	ZAAUUF_FU:46		FU:46	K	CHAR	10		User field 46
0050	ZAAUUF_FU:47		FU:47	K	CHAR	10		User field 47
0060	ZAAUUF_FU:48		FU:48	K	CHAR	10		User field 48
0070	ZAAUUF_FU:49		FU:49	K	CHAR	10		User field 49
0080	ZAAUUF_FU:50		FU:50	K	CHAR	10		User field 50
0090	ZAAUUF_FU:51		FU:51	K	CHAR	20		User field 51
0100	ZAAUUF_FU:52		FU:52	K	CHAR	20		User field 52
0110	ZAAUUF_FU:53		FU:53	K	CHAR	20		User field 53
0120	ZAAUUF_FU:54		FU:54	K	CHAR	20		User field 54
0130	ZAAUUF_FU:55		FU:55	K	CHAR	20		User field 55
0140	ZAAUUF_FU:56		FU:56	K	CHAR	20		User field 56
0150	ZAAUUF_FU:57		FU:57	K	CHAR	20		User field 57

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0160	ZAAUUF_FU:58		FU:58	K	CHAR	20		User field 58
0170	ZAAUUF_FU:59		FU:59	K	CHAR	20		User field 59
0180	ZAAUUF_FU:60		FU:60	K	CHAR	20		User field 60
0190	ZAAUUF_FU:61		FU:61	K	CHAR	20		User field 61
0200	ZAAUUF_FU:62		FU:62	K	CHAR	20		User field 62
0210	ZAAUUF_FU:63		FU:63	K	CHAR	20		User field 63
0220	ZAAUUF_FU:64		FU:64	K	CHAR	20		User field 64
0230	ZAAUUF_FU:65_1		FU:65_1	K	CHAR	30		User field 65
0240	ZAAUUF_FU:65_2		FU:65_2	K	CHAR	10		User field 65
0250	ZAAUUF_FU:66_1		FU:66_1	K	CHAR	30		User field 66
0260	ZAAUUF_FU:66_2		FU:66_2	K	CHAR	10		User field 66

Operation data

Operation data part one (ZOPER_01)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	PPPI_PHASE_SHORT_TEXT		AGBEZ_1	M	CHAR	30		Designation of the operation
0060	ZPPPI_PHASE_SHORT_TEXT		AGBEZ_2	M	CHAR	10		Designation of the operation
0070	PPPI_MATERIAL_NUMBER	MARA-BISMT	ATK	M	CHAR	18		Number of producing material; Article (number) Alphabetical characters in UPPER CASE

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0080	Z_PPPI_MATERIAL_TYPE_HYDRA	Characteristic in SAP	MATTYP	MM	CHAR	10		Material type of the article; if HYDRA MPL is used according to the configuration of the material type
0090	ZEXTPRIO	01-APRIO	EXTPRIO	M	CHAR	1		Priority
0100	PPPI_RESOURCE	03-ARBPL	MNR	M	CHAR	8		Planned work center; leer lassen, wenn der Arbeitsgang im PPS-System auf Gruppe geplant ist.
0110	PPPI_PLANT_OF_RESSOURCE		WERK:S	M	CHAR	4		Plant of the workcentre
0120	ZCOLOR	Classification characteristic on material level (header material)	COLOR	K	CHAR	20		Color of the material
0130	ZKST	Cost center linked to resource	KST	K	CHAR	8		Cost center; no processing in HYDRA (only information)
0140	ZKART		KART	K	CHAR	10		Cost type; no processing in HYDRA (only information)
0150	ZASTUFE	Fixed "2"	ASTUFE	K	CHAR	1		Authorization level for logging on/logging off OPs (lowest authorization = 1)
0160	ZRMNR	03-RUECK	RMNR	K	CHAR	10		Confirmation number
0170	ZDATTERMB	03-AFVV.FSAVD	DATTERMB	K	DATE	8		Scheduled start (date)
0180	ZZEITERMB	03-FSAVZ	ZEITERMB	K	TIME	6		Scheduled start (time) Please note: 240000 is not allowed; use instead 235959
0190	ZDATTERME	03-FSELD	DATTERME	K	DATE	8		Scheduled end (date)
0200	ZZEITERME	03-FSELZ	ZEITERME	K	TIME	6		Scheduled end (time) Please note: 240000 is not allowed; use instead 235959
0210	ZAGHD_DATFB	03-AFVV.FSAVD	DATFB	K	DATE	8		Earliest start (date) (AFVV.FSAVD)
0220	ZAGHD_ZEIFB	03-FSAVZ	ZEIFB	K	TIME	6		Earliest start (time) (AFVV.FSAVZ) Please note: 240000 is not allowed; use instead 235959
0230	ZAGHD_DATSB	03-SSAVD	DATSB	K	DATE	8		Latest start (date) (AFVV.SSAVD)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0240	ZAGHD_ZEISB	03-SSAVZ	ZEISB	K	TIME	6		Latest start (time) (AFVV.SSAVZ) Please note: 240000 is not allowed; use instead 235959
0250	ZAGHD_DATFE	03-FSELD	DATFE	K	DATE	8		Earliest end (date) (AFVV.FSELD)
0260	ZAGHD_ZEIFE	03-FSELZ	ZEIFE	K	TIME	6		Earliest end (time) (AFVV.FSELZ) Please note: 240000 is not allowed; use instead 235959
0270	ZAGHD_DATSE	03-SSELD	DATSE	K	DATE	8		Latest end (date) (AFVV.SSELD)
0280	ZAGHD_ZEISE	03-SSELZ	ZEISE	K	TIME	6		Latest end (time) (AFVV.SSELZ)
0290	ZAGHD_DATB		DATB	K	DATE	8		Planned start (date)
0300	ZAGHD_ZEIB		ZEIB	K	TIME	6		Planned start (time) Please note: 240000 is not allowed; use instead 235959
0310	ZAGHD_DATE		DATE	K	DATE	8		Planned end (date)
0320	ZAGHD_ZEIE		ZEIE	K	TIME	6		Planned end (time) Please note: 240000 is not allowed; use instead 235959
0330	ZAGHD_SGR:GUTB	Conversion from material master	SGR:GUTB	K	DEC	13	3	Target quantity (base quantity unit)
0340	ZAGHD_SGR:GUTP	03-MGVRG	SGR:GUTP	K	DEC	13	3	Target quantity (primary quantity unit)
0350	ZAGHD_SGR:GUTS	Conversion from material master	SGR:GUTS	K	DEC	13	3	Target quantity (secondary quantity unit)
0360	ZAGHD_SGR:GUTT	Conversion from material master	SGR:GUTT	K	DEC	13	3	Target quantity (tertiary quantity unit)
0370	ZAGHD_SGR:AUSB	Conversion from material master	SGR:AUSB	K	DEC	13	3	Target scrap (base quantity unit)
0380	ZAGHD_SGR:AUSP	03-ASVRG	SGR:AUSP	K	DEC	13	3	Target scrap (primary quantity unit)
0390	ZAGHD_SGR:AUSS	Conversion from material master	SGR:AUSS	K	DEC	13	3	Target scrap (secondary quantity unit)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0400	ZAGHD_SGR:AUST	Conversion from material master	SGR:AUST	K	DEC	13	3	Target scrap (tertiary quantity unit)
0410	ZAGHD_SGE:B	From material master MARA	SGE:B	K	CHAR	3		Base quantity unit
0420	ZAGHD_SGE:P	03-MEINH	SGE:P	K	CHAR	3		Primary quantity unit of entry
0430	ZAGHD_SGE:S	Conversion from material master	SGE:S	K	CHAR	3		Secondary quantity unit of entry
0440	ZAGHD_SGE:T	Conversion from material master	SGE:T	K	CHAR	3		Tertiary quantity unit of entry
0450	ZWEIGMENGE	03-MINWE	WEIGMENGE	K	DEC	13	3	Minimum send-ahead quantity (primary quantity unit)
0460	ZMENGEPROZ:UNTLI	AFPO-UNTTO	MENGEPROZ:UNTLI	K	DEC	13	3	Underdelivery in per cent
0470	ZOPT:UNTLI	V_TCORU-UNTLI for order type and plant	OPT:UNTLI	K	CHAR	1		Reaction to underdelivery " " No reaction "W" Warning "X" Error
0480	ZMENGEPROZ:UEBLI	AFPO-UEBTO	MENGEPROZ:UEBLI	K	DEC	13	3	Overdelivery in per cent
0490	ZOPT:UEBLI	V_TCORU-UEBLI for order type and plant	OPT:UEBLI	K	CHAR	1		Reaction to overdelivery " " No reaction "W" Warning "X" Error
0500	ZUMRFAKTP:Z	MARM	UMRFAKTP:Z	K	NUM	8		Factor for conversion primary quantity → base quantity (counter)
0510	ZUMRFAKTP:N	MARM	UMRFAKTP:N	K	NUM	8		Factor for conversion primary quantity → base quantity (denominator)
0520	ZUMRFAKTS:Z	MARM	UMRFAKTS:Z	K	NUM	8		Factor for conversion secondary quantity → base quantity (counter)
0530	ZUMRFAKTS:N	MARM	UMRFAKTS:N	K	NUM	8		Factor for conversion secondary quantity → base quantity (denominator)
0540	ZUMRFAKTT:Z	MARM	UMRFAKTT:Z	K	NUM	8		Factor for conversion tertiary quantity → base quantity (counter)
0550	ZUMRFAKTT:N	MARM	UMRFAKTT:N	K	NUM	8		Factor for conversion tertiary quantity → base quantity (denominator)

Operation data part two (ZOPER_02)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	ZRUEZ	03-RUEST	RUEZ	K	NUM	8		Setup time in seconds
0060	ZRUEZ:ZUSCHL		RUEZ:ZUSCHL	K	NUM	8		Addition of setup time in seconds
0070	ZBEARBZ	03-BEARZ	BEARBZ	K	NUM	8		Processing time in seconds
0080	ZPZ		PZ	K	NUM	8		Inspection time in seconds
0090	ZABRZ	03-ABRUE	ABRZ	K	NUM	8		Teardown time in seconds
0100	ZLIZ		LIZ	K	NUM	8		Delivery time in seconds
0110	ZFREMDFERT	T430-LIEF = X for control key of the phase in the order. Join AFVC with AFKO, field AUFPL	FREMDFERT	K	CHAR	1		External processing OP J/N
0120	ZRLZ:EXPR	Fix "RLFZ"	RLZ:EXPR	K	CHAR	6		Formula pro calculating the estimated remaining processing time (formula 1) Mainly relevant if the control center of HYDRA is used (HLS). Deviating settings are possible within the scope of the customizing of HYDRA.
0130	ZRLZ:EXPR2		RLZ:EXPR2	K	CHAR	6		Remaining running time (formula 2); optional (leave empty)
0140	ZVLZ		VLZ	K	NUM	8		Lead time in seconds
0150	ZLIEZ:MAX		LIEZ:MAX	K	NUM	8		Max. synchronization time in seconds
0160	ZWARTZ		WARTZ	K	NUM	8		Wait time in seconds
0170	ZWARTZ:MIN		WARTZ:MIN	K	NUM	8		Minimal waiting time in seconds
0180	ZLIEZ		LIEZ	K	NUM	8		Idle period in seconds
0190	ZLART		LART	K	CHAR	4		Wage type
0200	ZAKKORD		AKKORD	K	CHAR	1		Piece rate indicator/piece-rate premium

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0210	ZTE		TE	K	DEC	13	3	Premium specification te in seconds per 1000 pieces
0220	ZTR		TR	K	DEC	13	3	Premium specification tr in seconds
0230	ZTEB		TEB	K	DEC	13	3	Premium specification teb in seconds per 1000 pieces
0240	ZTRB		TRB	K	DEC	13	3	Premium specification trb in seconds
0250	ZVERARBCODE		VERARBCODE	M	CHAR	6		Processing code; fix "SYSTEM" Deviating settings are possible within the scope of the customizing of HYDRA.
0260	ZOPT:ERF	Fix "J"	OPT:ERF	M	CHAR	1		Can be entered J/N
0270	ZOPT:MULTIMNR		OPT:MULTIMNR	M	CHAR	1		Operation can be logged on parallel on different workplaces (J/N)
0280	ZOPT:CNR		OPT:CNR	MM	CHAR	1		Subject to batch tracing J/N
0290	ZOPT:SNR		OPT:SNR	M	CHAR	1		Subject to management in serial numbers J/N ("J" only relevant for ADE-SNR)
0300	ZSZY		SZY	K	NUM	8		Target cycle in seconds/1000, should be set; mandatory for MDE monitoring of the cycle of the machine
0310	ZTLG		TLG	K	NUM	8		Partitioning; should be pre-defined with 1; mandatory for MDE monitoring of the cycle of the machine
0320	ZIMPFAKT		IMPFAKT	K	DEC	13	3	Pulse factor; reserved; should be pre-defined with 1
0330	ZOPT:SPLIT		OPT:SPLIT	K	CHAR	1		Can be split J/N ("J" only relevant for ADE-SSG, ADE-SPL, HLS-SPL)
0340	ZMAXANZSPLIT		MAXANZSPLIT	K	NUM	8		Max. number of splits (only relevant if OPT:SPLIT = "J")
0350	ZMBVERH:RUE		MBVERH:RUE	K	DEC	5	2	Relation machines-user setting up; reserved.
0360	ZMBVERH:NORM		MBVERH:NORM	K	DEC	5	2	Relation machines-user producing;

Components of the operation

Components of the operation part one (ZCOMP_01)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	PPPI_MATERIAL_NUMBER	MARA-BISMT	ATK	M	CHAR	18		Material number; alphabetical characters in UPPER CASE
0060	PPPI_MATERIAL_SHORT_TEXT		ATKBEZ_1	K	CHAR	30		Material designation
0070	ZPPPI_MATERIAL_SHORT_TEXT		ATKBEZ_2	K	CHAR	10		Material designation
0080	ZAGKL_BEZ	RESB-POSNR	BEZ	K	CHAR	30		Comment 1
0090	ZAGKL_BEZ:2		BEZ:2	K	CHAR	30		Comment 2
0100	ZSLP	RESB-RSPOS	SLP	S	CHAR	10		Bill of material item / Position MPL-RF: Should position of the component in the layer structure (lamination)
0110	ZSLS		SLS	M	NUM	8		Level of bill of material Material components with level of bill of material > 1 are always saved in the material type "I" = info component
0120	ZART		ART	MM	CHAR	2		Material nature "M" (Consumable) material "T" Carrier material (only MPL-RF) "I" Info component
0130	Z_PPPI_MATERIAL_TYPE_HYDRA		MATTYP	MM	CHAR	10		MPL/MPL-RF: material type If HYDRA MPL is used a valid material type that is configured in HYDRA has to be indicated
0140	ZVERBR		VERBR	MM	CHAR	1		MPL/MPL-RF: Consumption type

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0150	ZOPT:ERSB		OPT:ERSB	MM	CHAR	1		MPL-RF: replaceable – may another material than the planned one be used for such a component? Only the same material type as the one of the material to be produced may be used then. Unplanned material could only log on at machine if correct machine status is set. MPL/MPL-RF: J/N Otherwise: Fix "N"
0160	ZOPT:WZW		OPT:WZW	MM	CHAR	1		MPL/MPL-RF: subject to changes; input batch change for a batch of this material forces an output batch change: if ART = "T" or "Z" → OPT:WZW must be "J" if ART = "I" or "A" → OPT:WZW must be "N" if ART = "M" → OPT:WZW: "J" or "N"
0170	ZAGKL_SGR:GUT		SGR:GUT	MM	DEC	13	3	MPL/MPL-RF: Required quantity for this component (position) in relation to the production of 1 unit/article in the primary quantity unit of the operation
0180	ZAGKL_SGE:GUT	RESB-EINHEIT	SGE:GUT	MM	CHAR	3		MPL/MPL-RF: quantity unit of the required quantity
0190	ZMENGEPROZ		MENGEPROZ	K	DEC	13	3	Required quantity in per cent Reserved, currently no use
0200	ZOTG		OTG	K	DEC	13	3	Upper tolerance limit in per cent; 3 places after decimal point Reserved; currently no use
0210	ZUTG		UTG	K	DEC	13	3	Lower tolerance limit in per cent; 3 places after decimal point Reserved; currently no use
0220	ZAGKL_EGR:GUT	RESB-ERFMG	EGR:GUT	K	DEC	13	3	Total quantity/requirement quantity: total quantity required for OP, i.e. for the quantity to be produced (output quantity)
0230	ZAGKL_EGE:GUT	RESB-EINHEIT	EGE:GUT	K	CHAR	3		Unit of the quantity required
0240	ZSLS:M		SLS:M	K	NUMC	8		Level of the bill of material of the mother material
0250	ZSLP:M		SLP:M	K	CHAR	10		Position of the bill of material of the mother material
0260	ZMENGE:FIX	RESB-FMENG	MENGE:FIX	K	CHAR	1		Indicator: fixed quantity No functionality in HYDRA, just displaying "J" Fixed Quantity " " no fixed quantity
0270	ZPPS:RETRO	RESB-RGEKZ	PPS:RETRO	K	CHAR	1		Indicator: Backflush (in the PPS) No functionality in HYDRA, just displaying

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0280	ZMATLISTINFO.LAGORT (:PPS)	RESB-LGORT	MATLISTINFO.LAGORT (:PPS)	K	CHAR	4		Storage location Will be mapped into FU:48 (MPDV internal)
0290	ZMATLISTINFO.LAGPZ (:SAP)	RESB-LGORT	MATLISTINFO.LAGPZ (:SAP)	K	CHAR	10		Storage bin Will be mapped into FU:49 (MPDV internal)
0300	ZAGKL_USRFLD		USRFLD	S	CHAR	8		User field key
0310	ZAGKL_FU:1		FU:1	K	DATE	8		User field 1
0320	ZAGKL_FU:2		FU:2	K	DATE	8		User field 2
0330	ZAGKL_FU:3		FU:3	K	DATE	8		User field 3
0340	ZAGKL_FU:4		FU:4	K	DATE	8		User field 4
0350	ZAGKL_FU:5		FU:5	K	DATE	8		User field 5
0360	ZAGKL_FU:6		FU:6	K	DATE	8		User field 6
0370	ZAGKL_FU:7		FU:7	K	NUM	8		User field 7
0380	ZAGKL_FU:8		FU:8	K	NUM	8		User field 8
0390	ZAGKL_FU:9		FU:9	K	NUM	8		User field 9
0400	ZAGKL_FU:10		FU:10	K	NUM	8		User field 10
0410	ZAGKL_FU:11		FU:11	K	NUM	8		User field 11
0420	ZAGKL_FU:12		FU:12	K	NUM	8		User field 12
0430	ZAGKL_FU:13		FU:13	K	NUM	8		User field 13
0440	ZAGKL_FU:14		FU:14	K	NUM	8		User field 14
0450	ZAGKL_FU:15		FU:15	K	NUM	8		User field 15
0460	ZAGKL_FU:16		FU:16	K	NUM	8		User field 16
0470	ZAGKL_FU:17		FU:17	K	NUM	8		User field 17
0480	ZAGKL_FU:18		FU:18	K	NUM	8		User field 18
0490	ZAGKL_FU:19		FU:19	K	NUM	8		User field 19
0500	ZAGKL_FU:20		FU:20	K	NUM	8		User field 20
0510	ZAGKL_FU:21		FU:21	K	NUM	8		User field 21
0520	ZAGKL_FU:22		FU:22	K	NUM	8		User field 22
0530	ZAGKL_FU:23		FU:23	K	DEC	13	3	User field 23
0540	ZAGKL_FU:24		FU:24	K	DEC	13	3	User field 24
0550	ZAGKL_FU:25		FU:25	K	DEC	13	3	User field 25

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0560	ZAGKL_FU:26		FU:26	K	DEC	13	3	User field 26
0570	ZAGKL_FU:27		FU:27	K	DEC	13	3	User field 27
0580	ZAGKL_FU:28		FU:28	K	DEC	13	3	User field 28

Components of the operation part two (ZCOMP_02)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	PPPI_MATERIAL_NUMBER	MARA-BISMT	ATK	M	CHAR	18		Material number; alphabetical characters in UPPER CASE
0060	ZSLP		SLP	S	CHAR	10		Bill of material item / Position MPL-RF: Should position of the component in the layer structure (lamination)
0070	ZAGKL_USRFLD		USRFLD	S	CHAR	8		User field key
0080	ZAGKL_FU:29	MARA-STOFF	FU:29	K	CHAR	1		User field 29
0090	ZAGKL_FU:30		FU:30	K	CHAR	1		User field 30
0100	ZAGKL_FU:31		FU:31	K	CHAR	1		User field 31
0110	ZAGKL_FU:32		FU:32	K	CHAR	1		User field 32
0120	ZAGKL_FU:33		FU:33	K	CHAR	1		User field 33
0130	ZAGKL_FU:34		FU:34	K	CHAR	1		User field 34
0140	ZAGKL_FU:35		FU:35	K	CHAR	1		User field 35
0150	ZAGKL_FU:36		FU:36	K	CHAR	1		User field 36
0160	ZAGKL_FU:37		FU:37	K	CHAR	1		User field 37
0170	ZAGKL_FU:38		FU:38	K	CHAR	1		User field 38
0180	ZAGKL_FU:39		FU:39	K	CHAR	1		User field 39

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0190	ZAGKL_FU:40		FU:40	K	CHAR	1		User field 40
0200	ZAGKL_FU:41		FU:41	K	CHAR	1		User field 41
0210	ZAGKL_FU:42		FU:42	K	CHAR	1		User field 42
0220	ZAGKL_FU:43		FU:43	K	CHAR	1		User field 43
0230	ZAGKL_FU:44		FU:44	K	CHAR	1		User field 44
0240	ZAGKL_FU:45	RESB-RSNUM	FU:45	K	CHAR	10		User field 45
0250	ZAGKL_FU:46		FU:46	K	CHAR	10		User field 46
0260	ZAGKL_FU:47	RESB-CHARG	FU:47	K	CHAR	10		User field 47
0270	ZAGKL_FU:48		FU:48	K	CHAR	10		User field 48
0280	ZAGKL_FU:49		FU:49	K	CHAR	10		User field 49
0290	ZAGKL_FU:50		FU:50	K	CHAR	10		User field 50
0300	ZAGKL_FU:51		FU:51	K	CHAR	20		User field 51
0310	ZAGKL_FU:52	RESB-MATNR	FU:52	K	CHAR	20		User field 52
0320	ZAGKL_FU:53		FU:53	K	CHAR	20		User field 53
0330	ZAGKL_FU:54		FU:54	K	CHAR	20		User field 54
0340	ZAGKL_FU:55		FU:55	K	CHAR	20		User field 55
0350	ZAGKL_FU:56		FU:56	K	CHAR	20		User field 56
0360	ZAGKL_FU:57		FU:57	K	CHAR	20		User field 57
0370	ZAGKL_FU:58		FU:58	K	CHAR	20		User field 58
0380	ZAGKL_FU:59		FU:59	K	CHAR	20		User field 59
0390	ZAGKL_FU:60		FU:60	K	CHAR	20		User field 60
0400	ZAGKL_FU:61		FU:61	K	CHAR	20		User field 61
0410	ZAGKL_FU:62		FU:62	K	CHAR	20		User field 62
0420	ZAGKL_FU:63		FU:63	K	CHAR	20		User field 63
0430	ZAGKL_FU:64		FU:64	K	CHAR	20		User field 64
0440	ZAGKL_FU:65_1		FU:65_1	K	CHAR	30		User field 65
0450	ZAGKL_FU:65_2		FU:65_2	K	CHAR	10		User field 65
0460	ZAGKL_FU:66_1		FU:66_1	K	CHAR	30		User field 66
0470	ZAGKL_FU:66_2		FU:66_2	K	CHAR	30		User field 66

Production resources and tools

Production resources and tools (ZAGFH_01)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	ZRESTYP		RESTYP	S	CHAR	4		Resource type; possible values: DNC DNC-program ENT Withdrawal unit TEM Temperature control unit VOR Fixture WNR Tool Further resource types (idents) can be defined in the configuration of the resource type (menu WRM: Basic data > Resource types) if the HYDRA tool and resource management (WRM) is used.
0060	PPPI_MATERIAL		ATK	S	CHAR	18		Resource number/material number; alphabetical characters UPPER CASE
0070	ZAGFH_ATKBEZ_1		ATKBEZ_1	K	CHAR	30		Designation – Part one
0080	ZAGFH_ATKBEZ_2		ATKBEZ_2	K	CHAR	10		Designation – Part two
0090	ZAGFH_BEZ		BEZ	K	CHAR	30		Comment 1
0100	ZAGFH_BEZ:2		BEZ:2	K	CHAR	30		Comment 2
0110	ZAGFH_SGR:GUT		SGR:GUT	M	QUAN	13	3	Required quantity
0120	ZAGFH_SGE:GUT		SGE:GUT	K	CHAR	3		Quantity unit
0130	ZAGKL_USRFLD	Fixed value	USRFLD	S	CHAR	8		User field key
0140	ZAGKL_FU:1		FU:1	K	DATE	8		User field 1
0150	ZAGKL_FU:2		FU:2	K	DATE	8		User field 2
0160	ZAGKL_FU:3		FU:3	K	DATE	8		User field 3
0170	ZAGKL_FU:4		FU:4	K	DATE	8		User field 4
0180	ZAGKL_FU:5		FU:5	K	DATE	8		User field 5

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0190	ZAGKL_FU:6		FU:6	K	DATE	8		User field 6
0200	ZAGKL_FU:7		FU:7	K	NUM	8		User field 7
0210	ZAGKL_FU:8		FU:8	K	NUM	8		User field 8
0220	ZAGKL_FU:9		FU:9	K	NUM	8		User field 9
0230	ZAGKL_FU:10		FU:10	K	NUM	8		User field 10
0240	ZAGKL_FU:11		FU:11	K	NUM	8		User field 11
0250	ZAGKL_FU:12		FU:12	K	NUM	8		User field 12
0260	ZAGKL_FU:13		FU:13	K	NUM	8		User field 13
0270	ZAGKL_FU:14		FU:14	K	NUM	8		User field 14
0280	ZAGKL_FU:15		FU:15	K	NUM	8		User field 15
0290	ZAGKL_FU:16		FU:16	K	NUM	8		User field 16
0300	ZAGKL_FU:17		FU:17	K	NUM	8		User field 17
0310	ZAGKL_FU:18		FU:18	K	NUM	8		User field 18
0320	ZAGKL_FU:19		FU:19	K	NUM	8		User field 19
0330	ZAGKL_FU:20		FU:20	K	NUM	8		User field 20
0340	ZAGKL_FU:21		FU:21	K	NUM	8		User field 21
0350	ZAGKL_FU:22		FU:22	K	NUM	8		User field 22
0360	ZAGKL_FU:23		FU:23	K	DEC	13	3	User field 23
0370	ZAGKL_FU:24		FU:24	K	DEC	13	3	User field 24
0380	ZAGKL_FU:25		FU:25	K	DEC	13	3	User field 25
0390	ZAGKL_FU:26		FU:26	K	DEC	13	3	User field 26
0400	ZAGKL_FU:27		FU:27	K	DEC	13	3	User field 27
0410	ZAGKL_FU:28		FU:28	K	DEC	13	3	User field 28

Production resources and tools (ZAGFH_02)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
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No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	ZRESTYP		RESTYP	S	CHAR	4		Resource type; possible values: DNC DNC-program ENT Withdrawal unit TEM Temperature control unit VOR Fixture WNR Tool Further resource types (idents) can be defined in the configuration of the resource type (menu WRM: Basic data > Resource types) if the HYDRA tool and resource management (WRM) is used.
0060	PPPI_MATERIAL		ATK	S	CHAR	18		Resource number/material number; alphabetical characters UPPER CASE
0070	ZAGKL_USRFLD	Fixed value	USRFLD	S	CHAR	8		User field key
0080	ZAGKL_FU:29		FU:29	K	CHAR	1		User field 29
0090	ZAGKL_FU:30		FU:30	K	CHAR	1		User field 30
0100	ZAGKL_FU:31		FU:31	K	CHAR	1		User field 31
0110	ZAGKL_FU:32		FU:32	K	CHAR	1		User field 32
0120	ZAGKL_FU:33		FU:33	K	CHAR	1		User field 33
0130	ZAGKL_FU:34		FU:34	K	CHAR	1		User field 34
0140	ZAGKL_FU:35		FU:35	K	CHAR	1		User field 35
0150	ZAGKL_FU:36		FU:36	K	CHAR	1		User field 36
0160	ZAGKL_FU:37		FU:37	K	CHAR	1		User field 37
0170	ZAGKL_FU:38		FU:38	K	CHAR	1		User field 38
0180	ZAGKL_FU:39		FU:39	K	CHAR	1		User field 39
0190	ZAGKL_FU:40		FU:40	K	CHAR	1		User field 40
0200	ZAGKL_FU:41		FU:41	K	CHAR	1		User field 41
0210	ZAGKL_FU:42		FU:42	K	CHAR	1		User field 42
0220	ZAGKL_FU:43		FU:43	K	CHAR	1		User field 43
0230	ZAGKL_FU:44		FU:44	K	CHAR	1		User field 44

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0240	ZAGKL_FU:45		FU:45	K	CHAR	10		User field 45
0250	ZAGKL_FU:46		FU:46	K	CHAR	10		User field 46
0260	ZAGKL_FU:47		FU:47	K	CHAR	10		User field 47
0270	ZAGKL_FU:48		FU:48	K	CHAR	10		User field 48
0280	ZAGKL_FU:49		FU:49	K	CHAR	10		User field 49
0290	ZAGKL_FU:50		FU:50	K	CHAR	10		User field 50
0300	ZAGKL_FU:51		FU:51	K	CHAR	20		User field 51
0310	ZAGKL_FU:52		FU:52	K	CHAR	20		User field 52
0320	ZAGKL_FU:53		FU:53	K	CHAR	20		User field 53
0330	ZAGKL_FU:54		FU:54	K	CHAR	20		User field 54
0340	ZAGKL_FU:55		FU:55	K	CHAR	20		User field 55
0350	ZAGKL_FU:56		FU:56	K	CHAR	20		User field 56
0360	ZAGKL_FU:57		FU:57	K	CHAR	20		User field 57
0370	ZAGKL_FU:58		FU:58	K	CHAR	20		User field 58
0380	ZAGKL_FU:59		FU:59	K	CHAR	20		User field 59
0390	ZAGKL_FU:60		FU:60	K	CHAR	20		User field 60
0400	ZAGKL_FU:61		FU:61	K	CHAR	20		User field 61
0410	ZAGKL_FU:62		FU:62	K	CHAR	20		User field 62
0420	ZAGKL_FU:63		FU:63	K	CHAR	20		User field 63
0430	ZAGKL_FU:64		FU:64	K	CHAR	20		User field 64
0440	ZAGKL_FU:65_1		FU:65_1	K	CHAR	30		User field 65
0450	ZAGKL_FU:65_2		FU:65_2	K	CHAR	10		User field 65
0460	ZAGKL_FU:66_1		FU:66_1	K	CHAR	30		User field 66
0470	ZAGKL_FU:66_2		FU:66_2	K	CHAR	30		User field 66

Documents of the operation (ZAGDC_01)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	ZAGDC_ATK		ATK	S	CHAR	18		Document ID: distinct key alphabetical characters UPPER CASE
0060	ZAGDC_ATKBEZ_1		ATKBEZ_1	M	CHAR	40		Designation – Part one
0060	ZAGDC_ATKBEZ_2		ATKBEZ_2	M	CHAR	40		Designation – Part two
0070	ZAGDC_BEZ		BEZ	K	CHAR	30		Comment 1
0080	ZAGDC_BEZ:2		BEZ:2	K	CHAR	30		Comment 2
0090	ZAGDC_PATH		PATH	M	CHAR	8		Reference to a mapping that is defined in the configuration of the mapping (menu file > System administration > Mappings).
0100	ZAGDC_DATEI_1		DATEI_1	M	CHAR	30		File name incl. file extension – Part one
0100	ZAGDC_DATEI_2		DATEI_2	M	CHAR	30		File name incl. file extension – Part two
0100	ZAGDC_DATEI_3		DATEI_3	M	CHAR	30		File name incl. file extension – Part three
0100	ZAGDC_DATEI_4		DATEI_4	M	CHAR	30		File name incl. file extension – Part four
0100	ZAGDC_DATEI_5		DATEI_5	M	CHAR	8		File name incl. file extension – Part five

Userfields of the operation

Userfields of the operation part one (ZUFIELD3)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	ZAGUF_USRFLD	Fixed value	USRFLD	S	CHAR	8		User field key
0060	ZAGUF_FU:1		FU:1	K	DATE	8		User field 1
0070	ZAGUF_FU:2		FU:2	K	DATE	8		User field 2
0080	ZAGUF_FU:3		FU:3	K	DATE	8		User field 3
0090	ZAGUF_FU:4		FU:4	K	DATE	8		User field 4
0100	ZAGUF_FU:5		FU:5	K	DATE	8		User field 5
0110	ZAGUF_FU:6		FU:6	K	DATE	8		User field 6
0120	ZAGUF_FU:7		FU:7	K	NUM	8		User field 7
0130	ZAGUF_FU:8		FU:8	K	NUM	8		User field 8
0140	ZAGUF_FU:9		FU:9	K	NUM	8		User field 9
0150	ZAGUF_FU:10		FU:10	K	NUM	8		User field 10
0160	ZAGUF_FU:11		FU:11	K	NUM	8		User field 11
0170	ZAGUF_FU:12		FU:12	K	NUM	8		User field 12
0180	ZAGUF_FU:13		FU:13	K	NUM	8		User field 13
0190	ZAGUF_FU:14		FU:14	K	NUM	8		User field 14
0200	ZAGUF_FU:15		FU:15	K	NUM	8		User field 15
0210	ZAGUF_FU:16		FU:16	K	NUM	8		User field 16
0220	ZAGUF_FU:17		FU:17	K	NUM	8		User field 17
0230	ZAGUF_FU:18		FU:18	K	NUM	8		User field 18
0240	ZAGUF_FU:19		FU:19	K	NUM	8		User field 19
0250	ZAGUF_FU:20		FU:20	K	NUM	8		User field 20
0260	ZAGUF_FU:21		FU:21	K	NUM	8		User field 21
0270	ZAGUF_FU:22		FU:22	K	NUM	8		User field 22
0280	ZAGUF_FU:23		FU:23	K	DEC	13	3	User field 23
0290	ZAGUF_FU:24		FU:24	K	DEC	13	3	User field 24
0300	ZAGUF_FU:25		FU:25	K	DEC	13	3	User field 25
0310	ZAGUF_FU:26		FU:26	K	DEC	13	3	User field 26

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0320	ZAGUF_FU:27		FU:27	K	DEC	13	3	User field 27
0330	ZAGUF_FU:28		FU:28	K	DEC	13	3	User field 28
0340	ZAGUF_FU:29		FU:29	K	CHAR	1		User field 29
0350	ZAGUF_FU:30		FU:30	K	CHAR	1		User field 30
0360	ZAGUF_FU:31		FU:31	K	CHAR	1		User field 31
0370	ZAGUF_FU:32		FU:32	K	CHAR	1		User field 32
0380	ZAGUF_FU:33		FU:33	K	CHAR	1		User field 33
0390	ZAGUF_FU:34		FU:34	K	CHAR	1		User field 34
0400	ZAGUF_FU:35		FU:35	K	CHAR	1		User field 35
0410	ZAGUF_FU:36		FU:36	K	CHAR	1		User field 36
0420	ZAGUF_FU:37		FU:37	K	CHAR	1		User field 37
0430	ZAGUF_FU:38		FU:38	K	CHAR	1		User field 38
0440	ZAGUF_FU:39		FU:39	K	CHAR	1		User field 39
0450	ZAGUF_FU:40		FU:40	K	CHAR	1		User field 40
0460	ZAGUF_FU:41		FU:41	K	CHAR	1		User field 41
0470	ZAGUF_FU:42		FU:42	K	CHAR	1		User field 42
0480	ZAGUF_FU:43		FU:43	K	CHAR	1		User field 43
0490	ZAGUF_FU:44		FU:44	K	CHAR	1		User field 44

Userfields of the operation part two (ZUFIELD4)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0050	ZAGUF_USRFLD		USRFLD	S	CHAR	8		User field key
0060	ZAGUF_FU:45		FU:45	K	CHAR	10		User field 45
0070	ZAGUF_FU:46		FU:46	K	CHAR	10		User field 46
0080	ZAGUF_FU:47		FU:47	K	CHAR	10		User field 47
0090	ZAGUF_FU:48		FU:48	K	CHAR	10		User field 48
0100	ZAGUF_FU:49		FU:49	K	CHAR	10		User field 49
0110	ZAGUF_FU:50		FU:50	K	CHAR	10		User field 50
0120	ZAGUF_FU:51		FU:51	K	CHAR	20		User field 51
0130	ZAGUF_FU:52 (PO_OUTPUT_COLOUR)		FU:52	K	CHAR	20		User field 52
0140	ZAGUF_FU:53 (PO_OUTPUT_MULTI_COLOUR)		FU:53	K	CHAR	20		User field 53
0150	ZAGUF_FU:54 (OP_OUTPUT_SHAPE)		FU:54	K	CHAR	20		User field 54
0160	ZAGUF_FU:55		FU:55	K	CHAR	20		User field 55
0170	ZAGUF_FU:56		FU:56	K	CHAR	20		User field 56
0180	ZAGUF_FU:57		FU:57	K	CHAR	20		User field 57
0190	ZAGUF_FU:58		FU:58	K	CHAR	20		User field 58
0200	ZAGUF_FU:59		FU:59	K	CHAR	20		User field 59
0210	ZAGUF_FU:60		FU:60	K	CHAR	20		User field 60
0220	ZAGUF_FU:61		FU:61	K	CHAR	20		User field 61
0230	ZAGUF_FU:62		FU:62	K	CHAR	20		User field 62
0240	ZAGUF_FU:63		FU:63	K	CHAR	20		User field 63
0250	ZAGUF_FU:64		FU:64	K	CHAR	20		User field 64
0260	ZAGUF_FU:65_1		FU:65_1	K	CHAR	30		User field 65
0270	ZAGUF_FU:65_2		FU:65_2	K	CHAR	10		User field 65
0280	ZAGUF_FU:66_1		FU:66_1	K	CHAR	30		User field 66
0290	ZAGUF_FU:66_2		FU:66_2	K	CHAR	10		User field 66

Additional information for coil based manufacturing (ZAGRF_01)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0010	PPPI_PROCESS_ORDER		AUNR	S	CHAR	12		Order number
0020	ZAFOLG	03-CY_SEQNR	AFOLG	S	NUMC	6		Sequence
0030	PPPI_OPERATION		AGNR	S	CHAR	4		Operation
0040	PPPI_PHASE		UVGNR	S	CHAR	4		Sub-operation
0050	ZRFAGTYP		RFAGTYP	K	CHAR	1		Flag → type of operation: " " No special processing "P" Packaging operation "M" material staging (customer specific) "G" glow-operation (customer specific) more (customer specific)
0060	ZRFABZ		RFABZ	MM	CHAR	1		Distinguishes mother and children OPs in case of a planned deduction. "M" Mother OP of a planned deduction "K" Child OP of a planned deduction (in this case a special material movement - 531 - is processed).
0070	ZRFOPT:RS		RFOPT:RS	MM	CHAR	1		Indicator for cutting coils (only relevant if it is a cutting OP) " " no cutting "T" cutting of coils active (numbering of daughter coil) "M" cutting active (mother coils are generated again)
0080	ZRFMANR		RFMANR	MM	CHAR	30		Cutting plan (order combination): The deduction (link) is assigned to the respective mother OP by means of this field and the following fields in case of a planned deduction. The mother OP references itself.
0090	ZRFTRANZ		RFTRANZ	MM	NUMC	5		In case of cutting operations: number of the planned daughter coils per cut.
0100	ZRFTRANZSUM		RFTRANZSUM	MM	NUMC	5		In case of cutting operations (mother OP): number of the planned daughter coils per cut (beyond all deductions/ operations).
0110	ZRFRANZ		RFRANZ	MM	NUMC	6		Planned number of coils (only information)
0120	ZRFSTKF		RFSTKF	MM	NUMC	8		Area of a piece Unit: MM2 / ST (integer)

No	Characteristic	SAP Table / Field	HYDRA Field	V	T	L	D	Description
0130	ZRFBSBRS		RFBSBRS	MM	DEC	10	3	Sum seam width (Sum "border width" If several coils are produced simultaneously in one series of operations this field contains the sum of the single seam widths. The seam width of the individual series of operations is explicitly set for each series of operations ("mother" and "child" series of operations) in case of deductions (no totals formation) Unit: MM
0140	ZRFBREITEE		RFBREITEE	MM	DEC	10	3	Input-width operation Unit: MM
0150	ZRFBREITEA	MARA-MTART	RFBREITEA	MM	DEC	10	3	Output-width operation Width of one outgoing roll within the operation (even if more than one roll is cut). The initial width of the individual operations is explicitly set for each operation ("mother" and "child" operations) in case of planned deductions (no totals formation). Unit: MM
0160	ZRFAGVFA		RFAGVFA	MM	DEC	10	3	Mass per unit area Unit: G/M2
0170	ZRFHUEGEW		RFHUEGEW	K	DEC	10	3	The weight of the casing of the daughter rolls is specified here in case of cutting processes

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4 PP-PI Confirmations HYDRA --> SAP

Record types and activities supported by HYDRA

HYDRA BDE sends time ticket related confirmation of the following record types to SAP R/3 PP:

Record type	SAP meaning	Triggering HYDRA action
00004	Time ticket partial confirmation	Start of an process order / operation (transferred for each start)
00004	Time ticket partial completion	Automatic or manual order interruption at the BDE terminal or console
00002	Time ticket completion	Message of a completed order at the BDE terminal or BDE console

Please note

If HYDRA MPL is active, whenever an output batch is created (output batch change message), in addition to the time tickets for OP interruption or logging off, an "0004" partial finish is created and transferred to SAP.

Data structure PI_PHCON

Message type)*: PI_PHCON

IDoc type)*: PI_PHCON

Message functions: -

Segments / categories: PI_PHCON

)* The information is used in HYDRA only for HYDRA-internal reasons

Characteristic	Data type	Description	Mand.
PPPI_ACTIVITY_1	Num (13) with 3 decimals	Activity to be confirmed 1	
PPPI_ACTIVITY_1_FINISHED	Char (1)	Remaining work for act. 1	
PPPI_ACTIVITY_1_UNIT	Char (6)	Unit for activity 1	
PPPI_ACTIVITY_2	Num (13) with 3 decimals	Activity to be confirmed 2	
PPPI_ACTIVITY_2_FINISHED	Char (1)	Remaining work for act. 2	
PPPI_ACTIVITY_2_UNIT	Char (6)	Unit for activity 2	

Characteristic	Data type	Description	Mand.
PPPI_ACTIVITY_3	Num (13) with 3 decimals	Activity to be confirmed 3	
PPPI_ACTIVITY_3_FINISHED	Char (1)	Remaining work for act. 3	
PPPI_ACTIVITY_3_UNIT	Char (6)	Unit for activity 3	
PPPI_ACTIVITY_4	Num (13) with 3 decimals	Activity to be confirmed 4	
PPPI_ACTIVITY_4_FINISHED	Char (1)	Remaining work for act. 4	
PPPI_ACTIVITY_4_UNIT	Char (6)	Unit for activity 4	
PPPI_ACTIVITY_5	Num (13) with 3 decimals	Activity to be confirmed 5	
PPPI_ACTIVITY_5_FINISHED	Char (1)	Remaining work for act. 5	
PPPI_ACTIVITY_5_UNIT	Char (6)	Unit for activity 5	
PPPI_ACTIVITY_6	Num (13) with 3 decimals	Activity to be confirmed 6	
PPPI_ACTIVITY_6_FINISHED	Char (1)	Remaining work for act. 6	
PPPI_ACTIVITY_6_UNIT	Char (6)	Unit for activity 6	
PPPI_CLEAR_RESERVATIONS	Char (1)	Clear reservations	
PPPI_CONFIRMATION_SHORT_TEXT	Char (30)	Confirmation text	
PPPI_EVENT_DATE	Date : DDMMYYYY	Date of event	X
PPPI_EVENT_TIME	Time : HHMMSS	Date/time of event	X
PPPI_OPERATION	Char (4)	Operation number	
PPPI_PHASE	Char (4)	Phase number	X
PPPI_PHASE_RESOURCE	Char (8)	Primary resource of a phase	
PPPI_PLANT_OF_RESOURCE	Char (4)	Plant of the resource	
PPPI_POSTING_DATE	Date : DDMMYYYY	Posting date	
PPPI_PROCESS_ORDER	Char (12)	Process order	X
PPPI_SCRAP_TO_CONFIRM	Num (13) with 3 decimals	Scrap to be confirmed	
PPPI_STATUS_CONFIRMED	Char (5)	Status f. activity confirmat.	
PPPI_UNIT_OF_MEASURE	Char (6)	Unit of measure	
PPPI_YIELD_TO_CONFIRM	Num (13) with 3 decimals	Yield to be confirmed	

PLEASE NOTE:

The confirmation of quantities ("00002" / "00004") via partial confirmations during simultaneous recording using total quantity counters on MDE machines is not possible, as the SAP system does not process negative quantities. This type of recording can lead to negative yield quantity bookings at the end of the operation.

This restriction does not apply, if such negative bookings can be processed (e.g. through additional use of the SAP standard BAPIs or customer specific processes).

5 MYERPRCK - Program Parameters

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Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

Parameters	Meaning/use	Relevant interfaces	Productive release
Program parameters to control processing:			
/MESTYP=XXXX	The parameter MESTYP defines the structure to be generated.	All	Yes
/GRP=XXXX	The grouping type specifies the criterion by which uploads should be grouped. Possible values: PLANT --> Groups by plant	Requires customizations	Requires customizations
/V=sssss	Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time. Use the parameter /V=sssss (sssss =	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	delay time in seconds) to activate the above described delay when the upload program is called. Examples: <pre>myerprck.exe/out /V=3600</pre> The system only uploads postings that are older than one hour.	SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	
/BIS=DDMMYYHHMM /BIS=HHMM /TILLDATE=MM/DD/YYYY /TILLTIME=sec after midnight	Use the parameter /BIS=DDMMYYHHMM (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day. <pre>Myerprck.exe /BIS=2505110600</pre> This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011. <pre>Myerprck.exe /BIS=0600</pre> This parameter uploads postings that were recorded until 06:00 a.m. of the current day.	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TZ=+/-sssss	Use the parameter /TZ=+/-sssss to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPDC interface according to its specifications.	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KST=XXX	Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center. Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /KST=BDE100 /KST=BDE200</pre> The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/CLEAR_RES	Use the parameter "/CLEAR_RES" to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.	SAP-PPDC	Yes
/NEG_MENGE	By default, quantities (L20/L40) cannot	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPPDCC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished. This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.		
/LA_MNR	The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM. The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type. Then the system uses the machine to identify the activity type from the activity types kept in HYDRA.	SAP-PMCC3	Yes
/IDENT_PRAEFIX=	In the upload structure of the SAP-PPPDCC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field. You can add a prefix to the EX_IDENT field to differentiate between uploads	SAP-PPPDCC SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	from various HYDRA subsystems connected to one SAP instance. <u>Example:</u> <pre>Myerprck.exe /IDENT_PRAEFIX=ABC</pre> The prefix may only include hexadecimal characters: A –H und 0 – 9.		
/ABZEICH=XX	While customizing the order type, you can specify that only signed data records are uploaded. Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TRANSFER=	Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/NOTTRANSFER=XXX	Use the parameter "/NOTTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SEK	The EIS-ERP interface uploads the times of resource performance accounts in hours. In particular with very short lead times this may effect that logon times are cut off by a conversion into hours. Use this program parameter to upload times in seconds.	EIS-ERP ESI-XPPS	Yes
/RMTYP=	When customizing the order type, you can assign an upload type to the order type. Use this program parameter to only upload data records of this upload type. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KAT=	When customizing the order type, you can connect the order type with a category. Use the program parameter /KAT= to only upload data records of this category. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SART=	The system only uploads ADE log postings of the specified record type. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the corresponding uploads when customizing the order type. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /SART=A /SART=E</pre> ⇒ The system only uploads A and E records.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/NOLOCK	When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,	All	Requires customizations

Parameters	Meaning/use	Relevant interfaces	Productive release
	the upload is not carried out. You can use this program parameter to prevent this check. Set this parameter, in particular, if the upload is not based on the table ade_protokoll.		
/EINH_CC34	The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter "/EINH_CC34" to upload the data in other units. The following units are supported: Hours: H, HUR, STD Minutes: MIN Seconds: SEC <u>Example:</u> <pre>Myerprck.exe /EINH_CC34=HUR</pre> The system uploads the recorded times in the unit "HUR" (hours).	SAP-PMCC3 SAP-PSCC4	Yes
/SDAT_STORNO	The SAP-PPPDCC interface transfers the change date along with the correction records. Use this program parameter to upload the initially collected shift date instead.	SAP-PPPDCC	Yes
/NORFC_STORNO	The SAP-PPPDCC interface transfers the cancellation records via sRFC. Use the program parameter to transfer the data in the IDoc format to SAP. To do	SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	so, inbound processing must be implemented in SAP. The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.		
/PI	If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly. Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).	SAP-PPDC SAP-PMCC3 SAP-PSCC4	Yes
/INDEX_TMP_TABLE	Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected. To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.	All	Requires customizations
/UE_PARAMS=	Program parameter for the stand-alone user exit processing (DD format).	Various	Yes
/NOSTORNO	Use this program parameter to prevent cancellation records from being uploaded. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the	All	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	corresponding uploads when customizing the order type.		
/RECALC_NEG_YIELD	Use this parameter to offset negative yield with already posted positive uploads.	SAP-PPPDCC	Requires customizations
Program parameters to use the SIGUSR communication:			
/LOGGING	Use this program parameter to activate communication from the database table HYD_LOGGING. To do so, a customization might be required.	INDIVIDUAL CASE	Yes
/WAIT_SIGUSR1=XX	The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.	INDIVIDUAL CASE	Yes
/PEEK_SIGUSR1=XX	Use this parameter to delay execution of an action triggered by the SIGUSR communication. The delay time is entered in seconds for this parameter. The program interprets this time as follows: If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds. If in a specific case, triggers would indeed arrive every second then the	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.		
/SEND_SIGUSR1=	This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication. Specify the process/program WITHOUT file extension.	INDIVIDUAL CASE	Yes
/COUNT_SIGUSR1=XX	Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance. Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1. Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	the DB. Benefits: If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.		
Program parameters for debugging/ tracing/ testing/ logging purposes:			
/ONLYERR	This program parameter specifies that system log entries are only created if an error occurred during uploading. This reduces the entries in the system log.	All	Yes
/SIM	The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to "True").	All	No
/SIMULATION	The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).	All	No

6 Application-relevant settings in HYDRA

Maintenance of the HYDRA distribution model - inbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA inbound processing:

Name of the parameter	Value
To process production orders	
Message type	PP_PI_PCS_HYDRA_INBOUND
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE =<MLE variant to use>
Description	PP-PI – Download of process orders
Log. Target system	Created logical system
Storage duration	10

Maintenance of the HYDRA distribution model - outbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA outbound processing:

Name of the parameter	Value
To upload phase confirmations	
Message type	PI_PHCON
Description	PP-PI – Upload of phase confirmations
IDoc-Typ	PI_PHCON
Storage duration	10
Log. Target system	Created logical system

Name of the parameter	Value
Segment name 1	PI_PHCON

Maintenance of the tRFC-Destination

Also for the MLE upload PP-PI-PCS specific settings have to be applied. In the configuration of [MLE Logical System](#) a new entry has to be applied for the program type "RFC-Client". In that entry the RFC destination name that has been created in SAP has to be provided.

Name of the parameter	Value
RFC_DESTINATION_FPR_TRFC	<Name of the RFC-destination created for HYDRA in SAP (SM59)>

Configuration of the confirmation prefix

In order to be able to transfer confirmations from several HYDRA systems back into SAP PP-PI, a unique identifier has to be applied. In case no INI-entry is given or the value is blank, the value "01" will be used.

This prefix can be configured using [HYDRA-INI-Configuration](#):

Parameter name	Value
INI-Name	PP-PI_PCS
Section	PP-PI-PCS_MSID_PREFIX
Key	KEY
Value	<Value to be used as unique identifier>
Active	Yes
Remark	PP-PI-PCS: Confirmation prefix

Maintenance of the HYDRA Scheduler

Use the [HYDRA Scheduler](#) to maintain entries for HYDRA outbound processing:

Parameter name	Value
To transfer confirmations from the application into the MLE outbound transactions	
Product key	SAP-PPPI
License key	SAP-PPPI
Command (Windows):	sh.exe ./myerprck.scr /MESTYP=PI_PHCON /KAT=FA
Command (Unix):	./myerprck.scr /MESTYP=PI_PHCON /KAT=FA
Comment:	SAP-PPPI: Confirmations → MLE outbound transactions
Intervall	5
To transfer confirmations from the MLE outbound transactions to SAP PP-PI	
Product key	SAP-PPPI
License key	SAP-PPPI
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=PI_PHCON /SINGLE_IDOC /SUBLEVEL=2 /SUBPROT=ON /PP-PI-PCS
Command (Unix):	./hysapupl.scr /UPLSEGNAM=BFLUSHDATAMTS /SINGLE_IDOC /SUBLEVEL=2 /SUBPROT=ON /PP-PI-PCS
Comment:	SAP-PPPI: Confirmations MLE outbound transactions → SAP
Intervall	5

7 Application-relevant customizing in SAP

Maintain control recipe destination

For each HYDRA system connected to SAP PP-PI a control recipe destination has to be maintained. In a first step the control recipe destination is linked with the RFC destination created in SAP transaction SM59. In further steps the process instructions have to be assigned.

To maintain the control recipe destination SAP transaction O10C is used (path with transaction SPRO: *Production Planning for Process Industries → Process Management → Decentralized Process Management → Define and Set Up Control Recipe Destinations*). It is important to maintain the control recipe destination of type “2”, as this is the type supported by the HYDRA PP-PI-PCS implementation.

Maintain control recipe destination in master recipe

The control recipe destination created in transaction O10C has to be assigned to a specific phase in the master recipe. By assigning control recipe destination that phase is relevant for downloading to HYDRA when the control recipe is created.

In most cases the operations in the master recipe are not relevant for HYDRA but only the phases.

Define background job for sending control recipe

In this step a background job for sending control recipes in the SAP client is defined. For each job, the following settings have to be made (SAP transaction SM36):

Start date of the job

One of the following options can be used:

The job should be started each time a new control recipe has been created in your client:

After event SAP_NEW_CONTROL_RECIPES

Event parameter <client>

Periodic job

The job should run periodically at certain intervals (option **Date/Time** with the time interval stored as a **period value**).

Steps to be carried out

The job should start the **ABAP report program** RCOCB006.

Define background job for sending process messages

In this step, a background job for sending process messages automatically is defined. This is necessary to process the process messages with confirmations sent from HYDRA to SAP within SAP and to post the data to the process order. To define the job SAP transaction SM36 is used.

Types of sending

The following options are available for background jobs:

Cross-plant sending

You use program RCOCB002 for this.

Plant-specific sending

You use program RCOCB004 for this. You specify the plant of the messages to be selected in a selection variant.

The system processes all messages that have status:

- Created
- To be resubmitted
- To be resubmitted with warning

Start date of the job

In certain time intervals

After the event SAP_NEW_PROCESS_MESSAGES, this means, every time new messages are available

For cross-plant sending, you specify the client as the event parameter.

For plant-specific sending, you specify the client and plant as the event parameters.

Define production scheduling profiles

In the production scheduling profile it is defined, if the control recipe is created automatically when releasing the process order. Production scheduling profiles can be assigned to a

- material (work scheduling screen in material master)
- production scheduler (Customizing)

The assignment to the material has a higher priority.

The production scheduling profile is copied to the production order or process order on order creation.

8 Supervise communication

Supervising in SAP

Communication SAP → HYDRA

The control recipes created in SAP and transferred to HYDRA can be supervised in SAP transaction CO53. The status of the control recipes is displayed as well as the date and time of creation.

Communication HYDRA → SAP

The process messages created in YHDAR and sent to SAP can be supervised in SAP transaction CO54. The status of the control recipe is displayed as well as the date and time of creation.

Supervising in HYDRA

The supervising in HYDRA can be done using [MLE inbound](#) and [MLE outbound transactions](#).

9 Protecting fields of planned operations

Purpose

Use the configuration described in this document to prevent specific data fields of a (planned) operation from being overwritten when the operation is transferred once more via the ERP interface.



This function only affects ANR.MODIFY and/or ANR.UPDATE and operations.

Operations are only updated if the status of the order/operation generally allows it. The configuration described below does not apply if the status (see [order status assignment](#)) cannot be changed in general.

Requirements

You require the relevant function authorization to access [INI configuration](#) and [INI data configuration](#).

Procedure from service pack 12 onwards (b_anr.dll version 8.1.1.354)

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	The value includes a condition. Enter the condition, for example, as follows: ANR.ATYP=AG
Active	Yes

Use "@" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



As of service pack 12 only use the "@" character as separator if you create new entries or change existing ones. You do not have to change existing configurations (prior to service pack 12). In this case, the "|" character is still supported.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

Procedure up to service pack 11

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	Enter the condition that has to be met to make sure fields will not be overwritten. Enter BAPI acronyms including value.
Active	Yes

Use "|" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.



Up to service pack 11 only use the "|" character as separator.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

If you cannot enter the pipe character ("|") using the GUI, you can still enter the values via the database:

- To do so, create a new entry as described above via the [INI configuration](#). Now use the following SQL statement to determine the internal DB counter for the header entry in the INI configuration:

```
select * from hyd_ini
```

- Determine the value of the "VERWEIS" column for the new entry.
- Create the required entries. Use the following SQL statement to assign the database table fields and application fields as described below:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung, aktiv)
values (<reference from previous SQL>, 'ANR', '<fields to be protected>', '<values>',
'<comment>', 'J')
```

Use the "|" (pipe) character to separate the acronyms of the fields you want to protect and the acronyms of the values.



Use a pipe character "|" to complete the list of the fields you want to protect and the list of values.

Database field	Values/content
INI_VERWEIS	The value of the VERWEIS column identified from the HYD_INI table via SQL.
SECTION	Section
IDENT	Key
VALUE	Value
BEMERKUNG (comment)	Comment
AKTIV	Active

List of frequently used acronyms

The following table lists the most frequently used acronyms and their meaning. Please contact MPDV Support if the list does not include the acronym you require.

Acronym	Meaning
ANR.MGRP	Machine group
ANR.MNR	Workplace/ machine
ANR.OPT:PLAN	Planning indicator: M Planned for workplace/machine G Planned for machine group
ANR.DATB	Start date planned (via HLS)
ANR.ZEIB	Start time planned (via HLS)
ANR.DATE	End date planned (via HLS)
ANR.ZEIE	End time planned (via HLS)

Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@', '<comment>',
'J')
```

Example: protect the start/end dates of a planned OP

If the operation is planned on a workstation and, as a result, its start time is specified, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@',
'<comment>', 'J')
```